

NIELIT O LEVEL • UNIT 04

Cascading Style Sheets (CSS)

Curriculum Hours: 25h (10h Theory + 15h Lab) | Topics: 8 | Language: English

UNIT OVERVIEW & PEDAGOGICAL SCOPE

Cascading Style Sheets (CSS) is the standardized stylesheet language used to describe the presentation, visual styling, formatting, and layout of documents written in HTML. Developed by H kon Wium Lie and Bert Bos in 1994 and standardized by W3C, CSS decouples presentation from document structure, enabling global site restyling by editing a single `.css` file. This unit covers CSS syntax rulesets, the 3 implementation methods (Inline, Internal, External), the cascading specificity hierarchy, CSS selectors (Universal, Type, Class, ID, Sub-selector, Attribute, Grouping, Pseudo-classes), CSS properties (Backgrounds, Typography, Box Model, Borders, Positioning modes: static, relative, absolute, fixed, sticky), styled CSS lists, styled data tables, navigation menu design (horizontal bars and dropdowns), and responsive image gallery design.

WHAT YOU WILL MASTER IN THIS UNIT

- The role and purpose of CSS in modern web design and the concept of 'Cascading'.
- The 3 methods of adding CSS: Inline (`style=""`), Internal (`<style>`), and External (`<link rel="stylesheet">`).
- CSS Selectors in detail: Universal (`*`), Type (`p`), Class (`.btn`), ID (`#header`), Sub-selector/Descendant (`div p`), Child (`div > p`), Attribute (`[type="text"]`), and Grouping (`h1, h2, h3`).
- CSS Properties: Background colors, gradients, background images, block formatting, line heights, and typography.
- The CSS Box Model: Content area, Padding, Border, Margin, and `box-sizing: border-box`.
- Positioning properties: `static` (normal flow), `relative`, `absolute`, `fixed`, and `sticky`.
- CSS Lists styling: `list-style-type`, `list-style-position`, and custom graphic markers.
- CSS Tables styling: `border-collapse`, zebra striping (`:nth-child(even)`), and cell padding.
- CSS Navigation Menu design: Horizontal flex navigation bars with hover state transitions and dropdown sub-menus.
- CSS Image Gallery: Responsive thumbnail grids with hover zoom transformations.

TOPIC 4.1: INTRODUCTION TO CSS

CSS stands for **Cascading Style Sheets**. It is a stylesheet language used to control the visual presentation, color schemes, typography, layout, and responsiveness of HTML web documents.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Before CSS, developers had to use ugly presentational HTML tags like `<font color="red">` on every single line. CSS separated content from styling, allowing one stylesheet to format thousands of pages simultaneously.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

A CSS ruleset consists of a **Selector** (which targets HTML elements) and a **Declaration Block** (enclosed in curly braces `{}`) containing one or more **Property: Value** pairs.

HTML/CSS/JS

```
/* CSS Rule Structure */
selector {
property: value;
property: value;
}

/* Concrete Example */
h1 {
color: #0056b3;
font-family: Arial, sans-serif;
font-size: 28px;
text-align: center;
}
```

ANATOMY OF A CSS RULESET

- Selector: Points to the HTML element you want to style (e.g. `h1`, `.card`, `#nav`).
- Declaration Block: Enclosed within opening `{` and closing `}` curly braces.
- Property: The specific visual style attribute being modified (e.g. `color`, `font-size`, `margin`).
- Value: The parameter assigned to the property (e.g. `blue`, `16px`, `center`), terminated by a semicolon `;`.
- Cascading Order: When conflicting rules exist, styles cascade based on Importance (`!important`), Specificity, and Source Order.

Imagine a 500-page university website. If the university changes its official brand color from blue to green: With CSS, the developer simply changes `color: green;` in one file (`style.css`), and all 500 pages update instantly.

- CSS stands for Cascading Style Sheets.
- CSS was created by H kon Wium Lie in 1994 and is standardized by the W3C.
- A CSS declaration always ends with a semicolon `;`, and declaration blocks are surrounded by curly braces `{}`.

QUICK TOPIC RECAP

CSS controls the visual presentation of HTML. A CSS rule consists of a Selector and a Declaration block containing `property: value;` pairs.

TOPIC 4.2: TYPES OF CSS (INLINE, INTERNAL, EXTERNAL)

There are three standardized methods to apply CSS styling rules to an HTML document: **Inline CSS**, **Internal CSS**, and **External CSS**.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Understanding the three methods and their cascading priority hierarchy is critical for writing maintainable code and avoiding styling conflicts.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Styles are declared either inside the HTML tag itself (`style` attribute), in the document `` (`<style>` tag), or in a standalone `.css` file linked via ``.

CASCADING SPECIFICITY HIERARCHY

- Priority Order (Lowest to Highest): 1. Browser Default Styles -> 2. External Stylesheet (`style.css`) -> 3. Internal Stylesheet (`<style>`) -> 4. Inline Styles (`style=""`) -> 5. `!important` rule.
- External CSS Advantages: Code reusability across entire website, complete separation of concerns, faster page loads (browser caches `.css` file after first visit).
- When to Use Inline CSS: Strictly for quick emergency testing or dynamic styling injected via JavaScript.

THE 3 METHODS OF APPLYING CSS COMPARED

Method / Type	Syntax Implementation	Scope & Maintenance	Cascading Priority Level
Inline CSS	<code><h1 style="color: red;">Title</h1></code>	Applies to that single individual element only; tedious to maintain	**Highest Priority** (overrides Internal and External)
Internal CSS	<code><head><style> h1 { color: red; } </style></head></code>	Applies to all matching elements within that single HTML file	Medium Priority (overrides External)
External CSS	<code><head><link rel="stylesheet" href="style.css"></head></code>	Applies globally to unlimited linked HTML pages; standard best practice	Base Priority (cached by browser; ultra-fast)

If `style.css` declares `p { color: black; }`, the `<style>` tag in `<head>` declares `p { color: blue; }`, and `<p style="color: red;">` is typed on the element: The paragraph displays in **RED** because Inline CSS has higher cascading priority than internal and external sheets.

- External CSS is the industry standard best practice, linked via `<link rel="stylesheet" href="style.css">`.
- Inline CSS has the highest cascading priority over Internal and External CSS.
- External CSS files are cached by browsers, dramatically improving website load speeds.

QUICK TOPIC RECAP

3 types of CSS: Inline (inside tag, highest priority), Internal (inside `<style>` in head), External (separate `.css` file via `<link>`, best practice).

TOPIC 4.3: CSS SELECTORS: UNIVERSAL, ID, TAG, CLASS, SUB SELECTOR, ATTRIBUTE, GROUP

A CSS Selector is the pattern expression at the start of a CSS rule that targets and selects specific HTML elements on the page to receive styling.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Mastering CSS selectors allows developers to precisely target individual buttons, specific form inputs, or entire layout sections without altering HTML markup.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

The browser's CSS matching engine parses selectors from right to left, matching elements against tag names, attributes, classes, and document tree hierarchies.

HTML/CSS/JS

```
/* Demonstrating various selectors */
* { box-sizing: border-box; } /* Universal */
h1, h2, h3 { color: #1a365d; } /* Grouping */
.alert-box { padding: 15px; } /* Class */
#main-nav { background: #333; } /* ID */
nav a { color: white; } /* Descendant */
input[type="submit"] { cursor: pointer; } /* Attribute */
```

MASTER CSS SELECTORS REFERENCE TABLE

Selector Type	CSS Selector Syntax	Target Description	Practical Code Example
Universal Selector	<code>`*`</code>	Selects every single element on the entire page	<code>* { margin: 0; padding: 0; }</code>
Type / Tag Selector	<code>`element`</code>	Selects all elements of that specific HTML tag name	<code>p { font-size: 16px; }</code>
Class Selector	<code>`classname`</code>	Selects elements sharing that class attribute (prefix with dot <code>`.`</code>)	<code>.btn { background: blue; color: white; }</code>
ID Selector	<code>`#idname`</code>	Selects the single unique element with that ID (prefix with hash <code>`#`</code>)	<code>#header { height: 80px; }</code>
Descendant (Sub Selector)	<code>`ancestor descendant`</code>	Selects elements nested anywhere inside the specified ancestor	<code>div p { color: gray; }</code>
Child Selector	<code>`parent > child`</code>	Selects only direct immediate children (not deeper grandchildren)	<code>ul > li { font-weight: bold; }</code>
Attribute Selector	<code>`[attribute='val']`</code>	Selects elements having that specific attribute and value	<code>input[type="text"] { border: 1px solid blue; }</code>
Grouping Selector	<code>`sel1, sel2, sel3`</code>	Applies identical styles to multiple selectors (separated by comma)	<code>h1, h2, h3 { font-family: sans-serif; }</code>

To style all submit buttons across a portal without affecting regular text boxes: Write ``input[type="submit"] { background: green; color: white; font-weight: bold; }``. Only submit buttons turn green.

- Class selectors start with a dot ``.`` (e.g. ``card``); ID selectors start with a hash ``#`` (e.g. ``header``).
- An ID must be unique within a webpage; a Class can be reused on multiple elements.
- The Universal selector is represented by an asterisk ``*``.
- Comma ``,`` groups selectors together; space represents descendant (sub-selector) nesting.

QUICK TOPIC RECAP

Selectors target HTML: ``*`` (all), ``tag`` (element), ``class`` (shared, dot), ``#id`` (unique, hash), ``parent child`` (descendant), ``[attr]`` (attribute), ``h1, h2`` (grouped).

TOPIC 4.4: CSS PROPERTIES: BACKGROUND, BLOCK, BOX, LIST, BORDER, POSITIONING PROPERTIES

CSS properties are the specific visual and structural attributes modified on target HTML elements. The **CSS Box Model** is the foundational layout architecture governing every element rendered on a webpage.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Without understanding the Box Model and positioning schemes, web pages suffer overlapping text, broken margins, and misaligned layouts.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Every HTML element is rendered by the browser as a rectangular box consisting of four concentric layers: 1. Content Area, 2. Padding, 3. Border, and 4. Margin.

HTML/CSS/JS

```

/* Practical Box Model & Positioning Example */
.notification-card {
  box-sizing: border-box;
  width: 320px;
  padding: 16px; /* Internal spacing */
  border: 1px solid #cbd5e1; /* Perimeter line */
  margin: 20px auto; /* External spacing & center */
  background-color: #ffffff;
  border-radius: 8px;
  position: relative;
}

.close-btn {
  position: absolute; /* Pinned to top-right of card */
  top: 8px;
  right: 8px;
  cursor: pointer;
}

```

THE 4 CONCENTRIC LAYERS OF THE CSS BOX MODEL

- 1. Content Area: The innermost core where text, images, or child elements reside (dimensions set by `width` and `height`).
- 2. Padding: Transparent internal clearance space between the content and the inner border (`padding: 10px;`).
- 3. Border: The perimeter boundary line wrapping around the padding and content (`border: 2px solid black;`).
- 4. Margin: Transparent external clearance space outside the border separating the element from neighboring elements (`margin: 15px;`).
- The `box-sizing: border-box` Rule: By default, padding and borders ADD to width. Setting `box-sizing: border-box` forces width to include padding and border, preventing layout overflow.

CORE CSS POSITIONING MODES EXPLAINED

Positioning Mode	Syntax	Behavior Relative To	Typical Practical Use Case
Static (Default)	<code>position: static;</code>	Normal document flow; top/left/right/bottom properties have NO effect	Default paragraphs and standard text flow
Relative	<code>position: relative;</code>	Its own normal original position (offsets without affecting neighbors)	Slight visual nudge; acts as anchor for absolute children
Absolute	<code>position: absolute;</code>	Nearest positioned ancestor (or viewport); removed from normal flow	Popup badges, modal close buttons, dropdown menus
Fixed	<code>position: fixed;</code>	The browser viewport window; remains pinned on screen during scroll	Sticky top navigation bars, floating 'Back to Top' buttons
Sticky	<code>position: sticky;</code>	Toggles between relative and fixed based on user's scroll position	Sticky table header rows, category sub-menus

To create a sticky header that stays at the top of the screen when users scroll down a 10-page article: Apply ``nav { position: fixed; top: 0; left: 0; width: 100%; z-index: 1000; }``. The navbar never scrolls away.

- The CSS Box Model consists of 4 parts from inside out: Content -> Padding -> Border -> Margin.
- `padding` is inside the border; `margin` is outside the border.
- `position: absolute` positions an element relative to its nearest positioned ancestor.
- `position: fixed` locks an element to the viewport window during scrolling.
- `box-sizing: border-box` includes padding and borders in element width calculations.

QUICK TOPIC RECAP

The Box Model: Content -> Padding -> Border -> Margin. Positioning: static (default), relative (offset from self), absolute (offset from ancestor), fixed (pinned to screen).

TOPIC 4.5: CSS LISTS

CSS list styling properties allow web developers to customize, position, or replace default bullet points and numbering markers in HTML lists (`<ul>` and `<ol>`).

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Default browser bullet discs look outdated. CSS list properties allow replacing bullets with custom icons or completely removing markers (`list-style: none;`) to construct navigation bars.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Developers apply `list-style-type`, `list-style-position`, and `list-style-image` to the list container.

HTML/CSS/JS

```
/* Resetting list for modern navigation bar */
ul.nav-menu {
  list-style-type: none; /* Removes bullet dots */
  margin: 0;
  padding: 0;
}

ul.nav-menu li {
  display: inline-block; /* Arranges items horizontally */
  margin-right: 20px;
}
```

CSS LIST PROPERTIES SPECIFICATION

CSS Property	Accepted Values	Visual Effect on List Markers
`list-style-type` (Unordered)	`none`, `disc`, `circle`, `square`	Controls bullet shape; `none` removes bullets entirely
`list-style-type` (Ordered)	`decimal`, `decimal-leading-zero`, `lower-roman`, `upper-roman`, `lower-alpha`	Controls numbering style (e.g. 01, 02, I, II, a, b)
`list-style-position`	`inside`, `outside` (default)	`outside` hangs markers outside text block; `inside` brings markers inside text flow
`list-style-image`	url('custom-bullet.png')	Replaces standard bullet dots with a custom miniature image
Shorthand `list-style`	list-style: square inside url('icon.png');	Combines type, position, and image in a single declaration

Every horizontal website navigation bar begins as an HTML `<ul>`. Applying `list-style-type: none; margin: 0; padding: 0;` strips away the bullet dots and indents. Adding `display: inline-block;` aligns the links horizontally across the header.

- `list-style-type: none;` is the foundational reset used to turn HTML lists into navigation menus.
- `list-style-position: inside` indents the bullet marker within the text content boundary.
- `list-style-image` allows substituting graphical icons for default bullets.

QUICK TOPIC RECAP

CSS lists: `list-style-type` sets marker shapes (square, roman) or removes them (`none`). `list-style-position` sets inside/outside. Used to build navigation menus.

TOPIC 4.6: CSS TABLES

CSS table styling properties format data tables by collapsing double borders, setting zebra-striped alternating rows, and adding hover highlights.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Raw HTML tables with `border="1"` look archaic with ugly separated double lines. Modern CSS creates clean, borderless, readable spreadsheet-grade tables.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

The `border-collapse: collapse;` property merges adjacent cell borders into a single sleek line. CSS pseudo-classes (`.nth-child(even)`) automate alternating row shading.

HTML/CSS/JS

```
/* Modern Clean CSS Table Styling */
table.styled-table {
  width: 100%;
  border-collapse: collapse; /* Merges double cell borders into one */
  font-family: Arial, sans-serif;
}

table.styled-table th,
table.styled-table td {
  padding: 12px 15px; /* Internal cell breathing room */
  border: 1px solid #e2e8f0;
  text-align: left;
}

table.styled-table th {
  background-color: #1e40af; /* Dark blue header */
  color: #ffffff;
  font-weight: bold;
}

table.styled-table tr:nth-child(even) {
  background-color: #f8fafc; /* Zebra striping on even rows */
}

table.styled-table tr:hover {
  background-color: #e2e8f0; /* Interactive row hover highlight */
}
```

KEY CSS TABLE PROPERTIES

- `border-collapse: collapse;`: Merges individual cell borders into a single clean border line (the most important table CSS property).
- `border-spacing: 10px;`: Sets gap distance between cells when `border-collapse` is set to `separate`.
- `:nth-child(even)` or `:nth-child(odd)`: Automatically applies light background shading to alternating rows (Zebra Striping) to prevent eye strain.
- `tr:hover`: Highlights the active row under the cursor with a subtle color change for improved readability.

A 50-row student marks table: Adding `tr:nth-child(even) { background-color: #f2f2f2; }` shades every second row grey. Students reading across wide columns can track grades without lines blurring together.

- `border-collapse: collapse;` collapses double borders into a single clean line.
- Zebra striping is achieved using `:nth-child(even)` or `:nth-child(odd)`.
- `text-align` and `vertical-align` control horizontal and vertical cell alignment.

QUICK TOPIC RECAP

CSS tables: Use ``border-collapse: collapse;`` to remove double borders, ``padding`` for cell spacing, and ``:nth-child(even)`` for alternating zebra striping.

TOPIC 4.7: CSS MENU DESIGN

CSS menu design involves styling an HTML unordered list (`<ul>` and `<li>`) into interactive horizontal or vertical navigation bars featuring hover states, active indicators, and dropdown sub-menus.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Navigation menus are the most important interactive component on a website, guiding visitors through site pages and services.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

List items are aligned horizontally using ``display: flex;`` or ``display: inline-block;``. Dropdown sub-menus are hidden with ``display: none;`` and revealed on hover using ``li:hover > ul { display: block; }``.

HTML/CSS/JS

```
<!-- CSS Dropdown Navigation Menu Example -->
<nav class="navbar">
  <ul class="nav-links">
    <li><a href="#home">Home</a></li>
    <li class="dropdown">
      <a href="#courses">Courses &#9662;</a>
      <ul class="dropdown-menu">
        <li><a href="#m1">M1-R5: IT Tools</a></li>
        <li><a href="#m2">M2-R5.1: Web Design</a></li>
        <li><a href="#m3">M3-R5: Python</a></li>
      </ul>
    </li>
    <li><a href="#contact">Contact</a></li>
  </ul>
</nav>

<style>
.navbar ul { list-style: none; margin: 0; padding: 0; background: #333; display: flex; }
.navbar a { display: block; padding: 14px 20px; color: white; text-decoration: none; }
.navbar a:hover { background: #555; }
.dropdown { position: relative; }
.dropdown-menu { display: none; position: absolute; top: 100%; left: 0; background: #444; min-width: 180px;
flex-direction: column; }
.dropdown:hover .dropdown-menu { display: flex; } /* Reveal dropdown on hover */
</style>
```

MENU DESIGN BEST PRACTICES

- Semantic Base: Always construct menus from semantic `<nav>` and `<ul>/<li>` tags for accessibility and SEO.
- Clickable Target Area: Set ``display: block;`` on `<a>` tags and add padding so the entire box area is clickable, not just the text characters.
- Visual Feedback: Always define ``:hover`` and ``:focus`` states so users receive instant visual confirmation of interactive elements.
- Mobile Hamburger Menu: On small screens, hide horizontal links and display a 3-bar hamburger icon (`☰`) to toggle a vertical slide-out drawer.

When you visit a portal and hover your mouse over 'Academics': A vertical box with 'Undergraduate', 'Postgraduate', and 'Syllabus' smoothly drops down. This is powered purely by CSS ``position: absolute;`` and ``:hover``.

- Navigation menus are built using semantic ``<nav>`` and ``<ul>`` lists.
- Dropdown menus hide child ``<ul>`` lists with ``display: none;`` and show them on hover via ``:hover``.
- Anchor ``<a>`` tags should have ``display: block;`` to enlarge the clickable target area.

QUICK TOPIC RECAP

CSS menus transform ``<ul>`` lists into horizontal bars using flexbox or inline-block. Dropdowns hide sub-menus (`display: none`) and reveal them on hover (`:hover`).

TOPIC 4.8: CSS IMAGE GALLERY

A CSS Image Gallery is a responsive visual grid layout designed to showcase multiple image thumbnails with borders, captions, hover zoom effects, and modal viewing capabilities.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Image galleries are essential for product catalogs, event photos, portfolio showcases, and tourism portals, requiring fluid responsiveness across mobile screens.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Images are organized in cards formatted with CSS Grid or Flexbox. CSS transitions and transformations (`transform: scale(1.05);`) add interactive hover micro-animations.

HTML/CSS/JS

```
/* Responsive CSS Image Gallery */
.gallery-container {
display: grid;
grid-template-columns: repeat(auto-fill, minmax(200px, 1fr));
gap: 16px;
padding: 16px;
}

.gallery-item {
border: 1px solid #ddd;
border-radius: 6px;
overflow: hidden;
box-shadow: 0 2px 4px rgba(0,0,0,0.1);
transition: transform 0.3s ease, box-shadow 0.3s ease;
}

.gallery-item:hover {
transform: scale(1.04); /* Smooth hover zoom effect */
box-shadow: 0 6px 12px rgba(0,0,0,0.2);
}

.gallery-item img {
width: 100%;
height: 150px;
object-fit: cover; /* Prevents image stretching */
display: block;
}

.gallery-caption {
padding: 10px;
text-align: center;
font-size: 14px;
background: #fff;
}
```

GALLERY DESIGN PRINCIPLES

- ``grid-template-columns: repeat(auto-fill, minmax(200px, 1fr));``: Creates an automated responsive grid that shows 4 columns on desktop and automatically wraps down to 1 column on phones.
- ``object-fit: cover;``: Scales images to completely fill thumbnail dimensions while preserving aspect ratio, preventing distorted or squished photos.
- ``transition`` and ``transform: scale();``: Produces smooth, elegant zoom animations when the mouse hovers over an image card.
- ``box-shadow``: Adds realistic paper elevation and depth.

A photography portfolio website: Photos are displayed in a clean 4-column grid. When a user hovers over a landscape photo, the card gently lifts upward with a soft drop shadow and zooms in slightly, inviting the user to click.

- ``object-fit: cover;`` prevents image distortion inside fixed-dimension gallery boxes.
- CSS Grid (``auto-fill``, ``minmax``) automatically reflows gallery columns across screen widths without manual media queries.
- ``transform: scale();`` creates subtle, modern hover micro-animations.

QUICK TOPIC RECAP

CSS image galleries organize photos into responsive grids using CSS Grid/Flexbox. ``object-fit: cover`` prevents stretching, and ``transform: scale();`` adds hover zoom.

TOPIC undefined:

- CSS (Cascading Style Sheets) separates presentation from HTML structure; rule syntax: ``selector { property: value; }``.
- The 3 types of CSS: Inline (``style=""``, highest priority), Internal (``<style>`` in ``<head>``), and External (``<link rel="stylesheet">``, industry best practice).
- CSS Selectors: Universal (``*``), Type (``p``), Class (`` .class``, dot), ID (``#id``, hash), Descendant (space), Child (``>``), Attribute (``[type="text"]``), Grouping (comma).
- The CSS Box Model consists of 4 concentric layers from inside out: Content -> Padding -> Border -> Margin.
- ``box-sizing: border-box`` forces element width calculations to include padding and borders.
- Positioning modes: ``static`` (default normal flow), ``relative`` (offset from itself), ``absolute`` (relative to positioned ancestor), ``fixed`` (locked to viewport), and ``sticky``.
- CSS Lists: ``list-style-type: none;`` strips default bullets, enabling the creation of navigation menus from ``<ul>`` tags.
- CSS Tables: ``border-collapse: collapse;`` merges double borders; ``:nth-child(even)`` automates zebra striping.
- CSS Menus: Built from semantic ``<nav>`` and ``<ul>``; dropdowns hide sub-menus (``display: none``) and reveal them on hover (``:hover``).
- CSS Galleries: Organize thumbnails using CSS Grid with ``object-fit: cover;`` to prevent distortion and ``transform: scale();`` for hover zoom effects.

KEY TERMS, ACRONYMS & OFFICIAL FULL FORMS